

$$4) \quad n=4, \quad \chi^2_{n-1} = \chi^2_3; \quad (2) \quad U_{1-\alpha/2} = F_{\chi^2_{n-1}}^{-1}(1-\alpha/2)$$

$$U_{\alpha/2} = F_{\chi^2_{n-1}}^{-1}(\alpha/2), \quad n-1=4-1=3$$

$$\alpha=0,05, \quad U_{1-\alpha/2} = U_{0,975} = 9,348 \text{ et } U_{\alpha/2} = U_{0,025} = 0,216$$

$$\hat{\sigma}_n^2 \quad IC_{1-\alpha}(\sigma^2) = \left[\frac{(n-1)\hat{\sigma}_n^2}{U_{1-\alpha/2}}, \frac{(n-1)\hat{\sigma}_n^2}{U_{\alpha/2}} \right]$$

$$IC_{0,95}(\sigma^2) = [15,08; 652,78]$$

$$\text{et } IC_{0,95}(\sigma) = [\sqrt{15,08}; \sqrt{652,78}] = [3,88; 25,55]$$

$$Ex 2 : 1) \quad \bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k; \quad \tilde{X} = n \bar{X}_n = \sum_{k=1}^n X_k$$

$$X_k \stackrel{\mathcal{L}}{\sim} \mathcal{B}_1(\pi) \quad \text{v.l.d} \Rightarrow \tilde{X}_n \stackrel{\mathcal{L}}{\sim} \mathcal{B}(n, \pi)$$

$$2) \quad E(\tilde{X}_n) = n\pi, \quad \sigma(\tilde{X}_n) = \sqrt{n\pi(1-\pi)}$$

$$\text{TCL} \quad \tilde{X}_n \xrightarrow{\mathcal{L}} \mathcal{N}(n\pi; \sqrt{n\pi(1-\pi)})$$

$$\text{et d'm} \quad \frac{\tilde{X}_n - E(\tilde{X}_n)}{\sigma(\tilde{X}_n)} = \frac{\tilde{X}_n - n\pi}{\sqrt{n\pi(1-\pi)}} \sim \mathcal{N}(0,1)$$

$$3) \quad z_{1-\alpha/2} = F^{*-1}(1-\alpha/2); \quad F^* = f^* \text{ la répartition de } \mathcal{N}(0,1).$$

$$\text{on a: } P\left(-z_{1-\alpha/2} \leq \tilde{X}_n^* = \frac{\tilde{X}_n - n\pi}{\sqrt{n\pi(1-\pi)}} \leq z_{1-\alpha/2}\right) = 1-\alpha$$

$$\Rightarrow P\left(\bar{X}_n - z_{1-\alpha/2} \frac{\sqrt{\pi(1-\pi)}}{\sqrt{n}} \leq \pi \leq \bar{X}_n + z_{1-\alpha/2} \frac{\sqrt{\pi(1-\pi)}}{\sqrt{n}}\right) = 1-\alpha$$